

University of Bedfordshire

<i>Unit Code:</i>	CIS046-3
<i>Unit Title</i>	Software for Enterprise
<i>Academic Year</i>	AY25/26
<i>Period of Study</i>	SEM2
<i>Date of Exam:</i>	<i>Exams Team Complete</i>
<i>Start Time:</i>	<i>Exams Team Complete</i>
<i>Time Allowed:</i>	Three Hours
<i>Unit Co-Ordinator:</i>	Syed Atif Moqurrab

Instructions for Candidates

- Answer ALL Questions
- Answers that are too verbose will receive lower marks. This helps to identify the use of software such as ChatGPT. Your answers must be as concise and focussed to the question as possible.
- Answers that are word-by-word identical to that of another student will automatically receive zero marks. Therefore, do not share your answers before the end of the exam.
- Answers that are a direct copy and paste from the Internet receive zero marks.
- Each Question is 10 marks except Question 8 which is 30 marks.
- Total: 100 marks.
- You need 40% = 40 marks to pass the exam.
- You need 70% = 70 marks for a first class.

Question 1 [10 Marks]

TechGear is a fast-growing tech retail company with 15 stores across the country. The company has recently experienced a surge in demand for its products, which has driven significant growth. However, with this growth, several operational challenges have emerged:

1. *TechGear* is encountering difficulties in ensuring that the right products are available at the right stores. Some stores frequently run out of high-demand items, while others have excess stock that takes up valuable space.
2. The company has a large customer base, but they are struggling to engage customers effectively. Customers have expressed dissatisfaction with generic promotions, and *TechGear* believes they need to better understand customer preferences and purchasing behaviour to improve loyalty and drive sales.

The CEO has asked for solutions to improve operational efficiency and enhance customer relationships as the company continues to grow.

Task:

1. **Identify the Two Key Business Areas:** Based on the case study, identify the two main business functions that *TechGear* should focus on improving to address their challenges.
2. **Recommend Enterprise Software:** For each identified business area, recommend a type of enterprise software solution that would help solve these challenges. Justify your recommendations by explaining how the software would improve operations.

Question 2 [10 marks]

TechSavvy, a startup developing an innovative e-commerce platform, has a team of developers working concurrently on multiple features. To manage the complexity and avoid conflicts in the rapidly evolving codebase, the team needs a robust version control system. Explain how Git and GitHub work together to enable effective version control in this scenario. In your answer, include:

- The basic Git workflow (staging, commits, branches).
- How GitHub facilitates collaboration (repositories, pull requests, forks).
- The benefits of using GitHub in a team environment.

Question 3 [10 marks]

Scenario:

FitTrack is a startup developing a cross-platform mobile application designed to offer personalized fitness tracking and health recommendations. The app must collect real-time data from various wearable devices and external health data services, such as heart rate, step count, sleep patterns, and calories burned, to generate customized workout and nutrition plans. Additionally, users can set specific goals and receive alerts based on their activity and health metrics. The app is intended to run on both Android and iOS platforms.

Task:

1. **Justify why an API is necessary** for the app to interact with external health and fitness data services.

2. **Determine the most appropriate type of API** (REST, SOAP, or GraphQL) for this app and justify your choice.
3. **Considering the need for cross-platform compatibility**, recommend tools or platforms that would support API integration while ensuring efficient performance on both Android and iOS. Discuss any potential limitations regarding the operating systems.

Question 4 [10 Marks]

Explain how **bias** can manifest in software development. In your answer, include:

- Different types of bias (e.g., algorithmic bias, UX/UI design bias).
- The consequences of biased software on end users.
- Best practices for identifying and mitigating bias in software development processes.

Question 5 [10 marks]

Evaluate the commercial implications of adopting service-based architectures in contemporary enterprises. In your answer, include:

- How service-based architectures are reshaping conventional business models.
- The financial benefits they offer, such as enhanced scalability and reduced operational costs.
- Examples of how service-based architectures can create additional revenue streams for organizations.

Question 6 [10 Marks]

Imagine you are tasked with designing the software architecture for a real-time grocery delivery platform, **FreshCart**, which processes thousands of orders and updates in real-time. You decide to implement an Event-Driven Architecture (EDA) for this system.

- **Explain** how event handlers, event listeners, and event queues would work together in FreshCart to manage real-time order tracking, inventory updates, and driver notifications. Describe the flow of events from when an order is placed until it is delivered and the inventory is updated.
- **Provide** a real-world example of how FreshCart could implement an Event-Driven Architecture to efficiently manage communication between customers, grocery stores, and delivery personnel during the order process. Discuss the benefits of using an EDA in this scenario.

Question 7 [10 marks]

Compare the **OSI Model** and the **TCP/IP Model** in terms of structure and functionality. In your answer:

- Explain the purpose of each model and how they are used in internet communication.
- Describe the layers of both models and their functions.
- Discuss the key differences and similarities between the OSI and TCP/IP models.

Question 8 [30 marks]

UrbanBrew Coffee House, a trendy chain known for its artisanal coffee, plans to roll out a customer loyalty program integrated into both its website and mobile app. Customers will be able to create accounts, track their purchase history, and earn points for every coffee bought either in-store or online. These points can later be redeemed for free beverages, merchandise, or discounts. Additionally, UrbanBrew aims to collect customer location data and behavioral insights to enhance their marketing strategies and tailor promotions. It is critical that the new loyalty feature maintains a consistent look and feel with UrbanBrew's current digital platforms. As the lead developer, your task is to design and implement these features while ensuring seamless integration with the existing systems.

a) With direct reference to the case study, explain in detail the role that HTML [4 marks], CSS [4 marks], JavaScript [4 marks], and PHP [4 marks] will play in implementing this loyalty program. Additionally, discuss whether Python could be used as an alternative to PHP [4 marks].

b) Discuss how sessions [2.5 marks], cookies [2.5 marks], IP addresses [2.5 marks], and passwords [2.5 marks] could be used to establish and manage online identity in this loyalty system.

END OF EXAM PAPER